

SCIKIT-LEARN REGRESSION MODELS

Hyperparameter Cheatsheet

1. Linear Models & Regularization

Linear Regression

Class: LinearRegression

Default parameters

- `fit_intercept = True`
- `positive = False`
- `copy_X = True`

GridSearch parameters

- `fit_intercept` $\in$ {True, False}
 - `positive` $\in$ {True, False}
-

Ridge Regression (L2)

Class: Ridge

Default parameters

- `alpha = 1.0`
- `fit_intercept = True`
- `solver = 'auto'`

GridSearch parameters

- `alpha` $\in$ {1e-3, 1e-2, 0.1, 1, 10, 100}
 - `solver` $\in$ {'auto', 'svd', 'cholesky', 'sag'}
 - `fit_intercept` $\in$ {True, False}
-

Lasso Regression (L1)

Class: Lasso

Default parameters

- `alpha = 1.0`
- `max_iter = 1000`
- `selection = 'cyclic'`

GridSearch parameters

- `alpha ∈ {1e-4, 1e-3, 1e-2, 0.1, 1}`
 - `max_iter ∈ {1000, 5000}`
-

ElasticNet (L1 + L2)

Class: ElasticNet

Default parameters

- `alpha = 1.0`
- `l1_ratio = 0.5`
- `max_iter = 1000`

GridSearch parameters

- `alpha ∈ {1e-3, 1e-2, 0.1, 1}`
 - `l1_ratio ∈ {0.1, 0.5, 0.9}`
-

2. Polynomial Regression

Polynomial Regression

Class: PolynomialFeatures + Linear model

Default parameters

- `degree = 2`
- `include_bias = True`
- `interaction_only = False`

GridSearch parameters

- `degree ∈ {2, 3, 4, 5}`
 - `interaction_only ∈ {True, False}`
-

3. Robust Regression (Outliers)

Huber Regression

Class: HuberRegressor

Default parameters

- `epsilon = 1.35`
- `alpha = 1e-4`

GridSearch parameters

- `epsilon ∈ {1.1, 1.35, 1.5, 2.0}`
 - `alpha ∈ {1e-4, 1e-3, 1e-2}`
-

RANSAC Regression

Class: RANSACRegressor

Default parameters

- `min_samples = None`
- `residual_threshold = None`

GridSearch parameters

- `min_samples ∈ {0.5, 0.7, 0.9}`
 - `residual_threshold ∈ {1.0, 2.0, 5.0}`
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4. Bayesian Regression

Bayesian Ridge Regression

Class: BayesianRidge

Default parameters

- `alpha_1 = 1e-6`
- `alpha_2 = 1e-6`
- `lambda_1 = 1e-6`
- `lambda_2 = 1e-6`

GridSearch parameters

- `alpha_1, alpha_2 ∈ {1e-6, 1e-5}`
 - `lambda_1, lambda_2 ∈ {1e-6, 1e-5}`
-

5. Support Vector Regression

Support Vector Regression (SVR)

Class: SVR

Default parameters

- `kernel = 'rbf'`
- `C = 1.0`
- `epsilon = 0.1`
- `gamma = 'scale'`

GridSearch parameters

- `kernel ∈ {'linear', 'rbf', 'poly'}`
 - `C ∈ {0.1, 1, 10, 100}`
 - `epsilon ∈ {0.01, 0.1, 0.5}`
 - `gamma ∈ {'scale', 'auto', 0.01, 0.1}`
-

6. Tree-based Regression

Decision Tree Regression

Class: DecisionTreeRegressor

Default parameters

- `max_depth = None`
- `min_samples_split = 2`

- `min_samples_leaf = 1`

GridSearch parameters

- `max_depth` $\in \{\text{None}, 5, 10, 20\}$
 - `min_samples_split` $\in \{2, 5, 10\}$
 - `min_samples_leaf` $\in \{1, 2, 5\}$
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Random Forest Regression

Class: RandomForestRegressor

Default parameters

- `n_estimators = 100`
- `max_depth = None`
- `max_features = 'sqrt'`

GridSearch parameters

- `n_estimators` $\in \{100, 300, 500\}$
 - `max_depth` $\in \{\text{None}, 10, 30\}$
 - `max_features` $\in \{'\text{sqrt}', '\text{log2}'\}$
 - `min_samples_leaf` $\in \{1, 2, 5\}$
-

7. k-Nearest Neighbors Regression

kNN Regression

Class: KNeighborsRegressor

Default parameters

- `n_neighbors = 5`
- `weights = 'uniform'`
- `p = 2` (*Euclidean distance*)

GridSearch parameters

- `n_neighbors` $\in \{3, 5, 7, 11\}$
- `weights` $\in \{'\text{uniform}', '\text{distance}'\}$
- `p` $\in \{1, 2\}$

8. Neural Network Regression

MLP Regression

Class: MLPRegressor

Default parameters

- `hidden_layer_sizes = (100,)`
- `activation = 'relu'`
- `alpha = 1e-4`
- `learning_rate_init = 0.001`

GridSearch parameters

- `hidden_layer_sizes` $\in \{(50,), (100,), (100,50)\}$
- `activation` $\in \{'relu', 'tanh'\}$
- `alpha` $\in \{1e-4, 1e-3\}$
- `learning_rate_init` $\in \{0.001, 0.01\}$

9. Gaussian Process Regression

Gaussian Process Regression (GPR)

Class: GaussianProcessRegressor

Default parameters

- `kernel = None`
- `alpha = 1e-10`
- `normalize_y = False`

GridSearch parameters

- `kernel` $\in \{RBF(), Matern()\}$
- `alpha` $\in \{1e-10, 1e-5\}$